

LAB ACTIVITY (Lab-04)

// 1. Write a Java program to check if a number is positive or negative using an if statement

```
import java.util.*;
public class positive
{
    public static void main(String[] args) {
        int num;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a number");
        num=sc.nextInt();

        if(num>=0){
            System.out.println("The number is positive");
        }
        else
            System.out.println("The number is negative");

    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>javac positive.java
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java positive
Enter number 1
23
The number is positive
```

// 2. Write a Java program to check if a number is even or odd using an if-else statement.

```
import java.util.*;
public class odd
{
    public static void main(String[] args) {
        int num;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a number");
        num=sc.nextInt();

        if(num%2==0){
            System.out.println("The number is Even");
        }
        else
            System.out.println("The number is Odd");

    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>javac odd.java
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java odd
Enter a number
45
The number is Odd
```

/* 3. Using if-else-if Ladder ,write a Java program to grade students based on their marks:

Marks >= 90: Grade A

Marks >= 80: Grade B

Marks >= 70: Grade C

Marks < 70: Fail */

```
import java.util.*;
public class ifelsemarks
{
    public static void main(String[] args) {
        int marks;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the marks");
        marks=sc.nextInt();

        if(marks>=90){
            System.out.println("Grade A");
        }
        else if(marks>=80)
            System.out.println("Grade B");
        else if(marks>=70)
            System.out.println("Grade C");
        else
            System.out.println("Fail");

    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>javac ifelsemarks.java
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java ifelsemarks
Enter the marks
67
Fail
```

/* 3. Using Switch statement, write a Java program to grade students based on their

marks:

Marks >= 90: Grade A

Marks >= 80: Grade B

Marks >= 70: Grade C

Marks < 70: Fail */

```
import java.util.*;
public class switchmarks
{
    public static void main(String[] args) {
        int marks;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the marks");
        marks=sc.nextInt();

        char grade;
        if(marks>=90){
            grade='A';
        }
        else if(marks>=80)
            grade='B';
        else if(marks>=70)
            grade='C';
        else
            grade='F';

        switch(grade){
            case 'A':
                System.out.println("Grade A");
                break;
            case 'B':
                System.out.println("Grade B");
                break;
            case 'C':
                System.out.println("Grade C");
                break;
```

```

        default:
            System.out.println("Fail");
            break;
    }

}
}

```

```

D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>javac switchmarks.java
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java switchmarks
Enter the marks
45
Fail

```

// 4. Write a Java program to check if a number is divisible by both 3 and 5

```

import java.util.*;
public class divisible
{
    public static void main(String[] args) {
        int num;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a number");
        num=sc.nextInt();

        if(num%3==0){
            if(num%5==0){
                System.out.println("The number is divisible by both 3 and 5");
            }
            else
                System.out.println("The number is divisible by 3 but not by 5");
        }
        else if(num%5==0){
            if(num%3==0){
                System.out.println("The number is divisible by both 3 and 5");
            }
            else
                System.out.println("The number is divisible by 5 but not by 3");
        }
    }
}

```

```

    }
}
}

```

```

D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java divisible
Enter a number
33
The number is divisible by 3 but not by 5

D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java divisible
Enter a number
25
The number is divisible by 5 but not by 3

```

/* 5. Write a Java program to print the season based on the month number:

12, 1, 2: Winter
3, 4, 5: Spring
6, 7, 8: Summer
9, 10, 11: Autumn */

```
import java.util.Scanner;
```

```
public class season {
```

```
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter the month number (1-12): ");
        int month = input.nextInt();
```

```
        String season;
```

```
switch (month) {  
    case 12:  
    case 1:  
    case 2:  
        season = "Winter";  
        break;  
    case 3:  
    case 4:  
    case 5:  
        season = "Spring";  
        break;  
    case 6:  
    case 7:  
    case 8:  
        season = "Summer";  
        break;  
    case 9:  
    case 10:  
    case 11:  
        season = "Autumn";  
        break;  
    default:  
        season = "Invalid month number";  
        break;  
}  
  
    System.out.println("The season for month " + month + " is: " + season);  
}  
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>javac season.java  
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java season  
Enter the month number (1-12): 7  
The season for month 7 is: Summer
```

/* 6. Write a Java program to calculate the discount on a product based on its price:

Price > 1000: 20% discount

Price > 500: 10% discount

Otherwise: 5% discount */

```
import java.util.Scanner;
public class discount {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the product price: ");
        double price = sc.nextDouble();

        double discountPercentage;

        if (price > 1000) {
            discountPercentage = 0.20;
        } else if (price > 500) {
            discountPercentage = 0.10;
        } else {
            discountPercentage = 0.05;
        }

        double discountAmount = price * discountPercentage;
        double finalPrice = price - discountAmount;

        System.out.println("Original Price: " + String.format("%.2f", price));
        System.out.println("Discount Applied: " + (discountPercentage * 100) +
            "%");
        System.out.println("Discount Amount: " + String.format("%.2f",
            discountAmount));
        System.out.println("Final Price after Discount: " + String.format("%.2f",
            finalPrice));

    }
}
```

}

```
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>javac discount.java
D:\SEMESTER I\LAB WORK\Java Programming\lab3(3008)>java discount
Enter the product price: 2000
Original Price: 2000.00
Discount Applied: 20.0%
Discount Amount: 400.00
Final Price after Discount: 1600.00
```